THE COPPERBELT UNIVERSITY

SCHOOL OF INFORMATION COMMUNICATION TECHNOLOGY (SICT)

CS301 – PROJECT MANAGEMENT

LITERATURE REVIEW

PROJECT NAME: TAX-OPOLOGY: THE PUBLIC GOOD GAME

GROUP NUMBER: 12

PROGRAMME: COMPUTER SCIENCE

YEAR OF STUDY: 3

PROJECT SUPERVISOR: MR. MUMBA

SIN	FIRST NAME	LAST NAME
23124656	ISAAC	TEMBO
23122916	DARIUS	CHUNGU
23138696	BENJAMIN	CHAAMBWA
23127482	LWANGA	MWABA

Approved:

____

Contents

1.1 INTRODUCTION

This document presents a comprehensive review of literature relevant to the development of "Tax-opoly: The Public Good Game," a digital educational tool designed to improve understanding of taxation and public goods. The review draws upon multiple disciplines, including tax compliance theory, behavioral economics, game-based learning, gamification, and existing educational applications. The purpose of this review is to establish the theoretical foundation for the project, identify gaps in current knowledge and practice, and justify the proposed approach to developing an interactive game that models tax compliance as a strategic choice with communal outcomes.

The document is structured as follows: Section 1.2 examines theoretical literature on tax compliance, exploring models that explain why citizens pay or evade taxes and the role of education in shaping compliance behaviour. Section 1.3 reviews behavioral economics research on public goods and collective action, including the classic Public Goods Game and its implications for understanding cooperation. Section 1.4 explores the principles of game-based learning and gamification, providing evidence for why games are effective educational tools. Section 1.5 analyses existing applications similar to Tax-opoly, including commercial games, educational simulations, and gamified learning platforms, evaluating their strengths and limitations. Section 1.6 stresses the lessons learnt from the literature, and Section 1.7 offers a critical critique of the reviewed work, identifying gaps that Tax-opoly aims to address. Finally, section 1.8 concludes the chapter with a summary of key findings and their implications for this project.

1.2 THEORETICAL FOUNDATIONS OF TAX COMPLIANCE

1.2.1 Economic Models of Tax Compliance

The study of tax compliance has long been dominated by economic models that view taxpayers as rational actors making calculated decisions about whether to comply with tax obligations. The foundational work in this area is the economic deterrence model developed by Allingham and Sandmo (1972), which applies Becker's (1968) economics-of-crime framework to tax evasion. According to this model, individuals decide whether to evade taxes by weighing the expected benefits of evasion against the potential costs of detection and punishment. The key variables in this calculation include the probability of audit, the penalty rate if caught, and the individual's risk preferences.

Allingham and Sandmo's (1972) model predict that compliance increases when audit probabilities and penalty rates are higher. However, empirical research has consistently shown that actual compliance rates are higher than what economic models would predict based solely on deterrence factors (Alm, McClelland, & Schulze, 1992). This discrepancy, often called the "compliance puzzle," suggests that non-economic factors play a significant role in taxpaying behavior. As Kirchler (2007) observes, if taxpayers were purely rational economic actors, evasion would be far more widespread than it actually is, given the relatively low audit rates in most countries.

1.2.2 Behavioural and Psychological Approaches

Recognising the limitations of purely economic models, researchers have increasingly turned to behavioural and psychological explanations of tax compliance. Kirchler (2007) proposed the "slippery slope framework," which distinguishes between voluntary compliance based on trust in authorities and enforced compliance based on the power of authorities. According to this framework, when citizens trust that government uses tax revenues wisely and fairly, they are more likely to comply voluntarily. Conversely, when trust is low, authorities must rely on enforcement, which can be costly and may breed resistance.

The concept of tax morale has emerged as a crucial factor in understanding compliance behaviour. Torgler (2007) defines tax morale as the intrinsic motivation to pay taxes, shaped by social norms, perceptions of fairness, and civic duty. Studies across multiple countries have found that tax morale varies significantly and correlates strongly with actual compliance behaviour (Alm & Torgler, 2006). In the Zambian context, ActionAid Zambia (n.d.) notes that perceptions of government transparency and the equitable use of tax revenues significantly influence citizens' willingness to comply with tax obligations.

Fiscal transparency emerges as a key theme in the literature. Balaguer-Coll et al. (2019) demonstrate that when citizens have access to clear information about how public funds are allocated and spent, their trust in government increases, and their attitudes toward taxation become more positive. This finding has direct implications for educational interventions: tools that make the link between tax payments and public goods visible and understandable may enhance tax morale and ultimately improve compliance.

1.2.3 Tax Education and Financial Literacy

The role of education in improving tax compliance has received increasing attention from researchers and policymakers. The Commonwealth Secretariat (2017) emphasises that financial literacy programmes, including those focused on taxation, are essential for inclusive economic growth. In developing economies particularly, where informal sectors are large and formal tax systems may be poorly understood, educational interventions can play a crucial role in building a culture of compliance.

Siregar and Suryanto (2024) review gamified learning environments in financial education and find that interactive, engaging approaches are significantly more effective than traditional instructional methods at improving both knowledge retention and attitudes toward financial topics. Their research supports the proposition that game-based learning could be particularly valuable for tax education, a domain often perceived as dry or intimidating by learners.

The Zambia Revenue Authority (2023) reports that despite ongoing public awareness campaigns, many Zambian citizens remain unclear about how tax revenues are used and why compliance matters. This knowledge gap contributes to the revenue shortfalls that hinder investment in public services, creating a cycle of underdevelopment and mistrust. The Authority's findings underscore the need for innovative educational approaches that can reach diverse audiences and make abstract fiscal concepts tangible and personally relevant.

1.3 PUBLIC GOODS AND COLLECTIVE ACTION

1.3.1 The Public Goods Problem in Economic Theory

Public goods are defined in economic theory as goods that are non-excludable and non-rivalrous, meaning that individuals cannot be effectively excluded from using them, and one person's use does not reduce availability to others (Samuelson, 1954). Classic examples include clean air, national defense, and, in the context of this project, publicly funded services such as roads, schools, and healthcare. The defining challenge of public goods provision is the "free rider problem": because individuals can benefit from public goods regardless of whether they contribute to their provision, rational actors may choose not to contribute, hoping that others will bear the cost (Olson, 1965).

This collective action dilemma lies at the heart of taxation. Taxes are compulsory contributions designed to overcome the free rider problem and ensure that public goods can be funded. However, when citizens do not understand or accept this logic, they may resist taxation, viewing it as an unjustified taking of private resources rather than an investment in shared prosperity.

1.3.2 Experimental Research on Public Goods

Behavioural economists have extensively studied cooperation and free riding using the Public Goods Game, a laboratory experiment that simulates collective action dilemmas. In a standard Public Goods Game, participants receive an initial endowment and privately decide how much to contribute to a shared pool. The total contributions are multiplied by a factor greater than one (reflecting the social returns to cooperation) and then distributed equally among all participants, regardless of their individual contributions (Ledyard, 1995).

Fehr and Gächter (2000) conducted seminal research on the Public Goods Game, demonstrating that while free riding is common, so too is cooperation, and that the opportunity to punish free riders can sustain high levels of contribution. Their findings reveal that many individuals are "conditional cooperators" who contribute when they believe others will also contribute, and that social norms and the visibility of others' actions significantly influence behaviour. However, as the authors note, the abstract nature of laboratory experiments may limit their accessibility and impact for general audiences.

The CORE Economics Team (2020) incorporates public goods experiments into their introductory economics curriculum, demonstrating the pedagogical value of interactive demonstrations.

Students who participate in such experiments show deeper understanding of collective action problems compared to those who only read about them. This finding supports the potential of interactive tools like Tax-opoly to enhance learning about public goods.

1.3.3 Taxation as a Public Goods Contribution

Viewing tax compliance through the lens of public goods theory reframes the issue from one of individual obligation to one of collective investment. When citizens pay taxes, they are contributing to a shared pool that funds services from which everyone benefits. However, this perspective is not automatically intuitive, particularly in contexts where public services are inadequate or where corruption undermines trust that contributions will be used effectively.

Kirchler (2007) argues that fostering a "tax climate" in which compliance is seen as a social norm rather than a burden requires that citizens perceive the link between their contributions and visible public goods. Educational interventions that make this link explicit and demonstrate the multiplier effects of collective investment could therefore play a crucial role in building tax morale.

1.4 GAME-BASED LEARNING AND GAMIFICATION

1.4.1 Theoretical Foundations of Game-Based Learning

Game-based learning rests on the premise that games create powerful learning environments by engaging players in active problem-solving, providing immediate feedback, and allowing experimentation in low-stakes contexts. Gee (2003), in his influential work "What Video Games Have to Teach Us About Learning and Literacy," argues that well-designed games embody principles that cognitive science has identified as crucial for effective learning. These include: allowing players to learn through failure without real-world consequences; providing information just in time, when it is needed; and enabling players to understand complex systems by interacting with simplified models.

Prensky (2001) similarly contends that digital game-based learning aligns with the learning preferences of contemporary generations, who have grown up with interactive media and expect engagement rather than passive reception of information. While Prensky's generational arguments have been critiqued as oversimplified, the underlying point about the motivational power of interactive learning remains widely accepted.

1.4.2 Empirical Evidence for Game-Based Learning Effectiveness

A growing body of empirical research supports the effectiveness of game-based learning across multiple domains. Wouters et al. (2013) conducted a meta-analysis of 39 studies comparing serious games to conventional instructional methods. They found that serious games were more effective for learning and retention, and that they generated greater learner motivation. However, the authors also noted that the advantages of games were enhanced when they were used as supplements to other instruction rather than replacements, and when they incorporated multiple sessions rather than single exposures.

In the domain of civic and economic education specifically, researchers have found promising results. Hsu, Chang, and Hung (2018) investigated the use of a simulation game to teach economic concepts and found significant improvements in both knowledge and attitudes compared to traditional instruction. Students who played the game demonstrated better understanding of abstract concepts such as supply and demand, and reported greater interest in economics as a subject.

1.4.3 Gamification: Principles and Applications

Gamification refers to the application of game design elements in non-game contexts (Deterding et al., 2011). Unlike serious games, which are full-fledged games designed for educational purposes, gamification involves incorporating game-like features such as points, badges, leaderboards, and progress tracking into non-game environments to increase engagement and motivation.

Deterding et al. (2011) emphasise that effective gamification requires meaningful integration of game elements with the underlying activity. Simply adding points or badges to a task does not automatically enhance motivation; rather, game elements should be designed to support intrinsic motivation by providing feedback, acknowledging progress, and creating a sense of competence and autonomy.

Seaborn and Fels (2015) conducted a systematic review of gamification research, finding mixed evidence for its effectiveness. While many studies reported positive outcomes, the authors noted that poorly designed gamification could actually reduce motivation, particularly if it created extrinsic pressures that undermined intrinsic interest. This finding highlights the importance of thoughtful design in educational games like Tax-opoly: game mechanics must be aligned with learning objectives and should support rather than distract from educational goals.

1.4.4 Applications to Financial and Civic Education

The application of game-based learning and gamification to financial and civic education has gained momentum in recent years. The Organisation for Economic Co-operation and Development (OECD, 2020) has developed Taxland, a digital game designed to teach tax compliance and civic responsibility to young audiences. While Taxland represents an important step forward, its relatively simple mechanics and limited strategic depth may limit its effectiveness for older or more sophisticated learners.

Siregar and Suryanto (2024) review recent developments in gamified financial education and conclude that the most effective interventions combine clear learning objectives with engaging gameplay mechanics that allow players to experiment with decisions and observe consequences. Their research supports the design approach proposed for Tax-opoly, in which players make strategic choices about tax compliance and see the resulting impacts on community quality of life.

1.5 ANALYSIS OF SIMILAR APPLICATIONS

1.5.1 Monopoly (Hasbro, 1935)

Monopoly is perhaps the world's most famous board game, having sold hundreds of millions of copies since its introduction during the Great Depression (Hasbro, 2019). The game simulates real-estate markets, with players buying, renting, and selling properties with the goal of bankrupting opponents. Its enduring popularity demonstrates the appeal of economic simulation games and their ability to engage players with concepts such as asset management, risk assessment, and negotiation.

However, Monopoly's design embodies a distinctly individualistic, zero-sum worldview. Players succeed by accumulating wealth at others' expense, and there is no mechanism for collective investment or shared prosperity. As Pilon (2015) notes in her history of the game's origins, Monopoly was actually derived from an earlier game, "The Landlord's Game," designed by Lizzie

Magie to demonstrate the injustices of concentrated land ownership. The commercial version stripped away this critical perspective, leaving only competitive accumulation.

For the purposes of tax education, Monopoly's limitations are significant. The game does not model taxation as a public good, nor does it illustrate the benefits of collective investment. Players learn to maximise individual gain but receive no exposure to concepts of civic contribution or shared outcomes. Tax-opoly builds upon Monopoly's familiar mechanics while fundamentally reorienting the game's objectives toward cooperation and communal wellbeing.

1.5.2 Public Goods Game (Experimental Economics)

The Public Goods Game, in its various forms, has been a workhorse of experimental economics for decades (Ledyard, 1995). In laboratory settings, researchers use the game to study how factors such as communication, punishment, and group identity affect cooperation. The game's strength lies in its clean, abstract representation of the collective action dilemma, which allows researchers to isolate specific variables and test theoretical predictions.

Despite its scientific value, the Public Goods Game has significant limitations as an educational tool for general audiences. Its abstract presentation, typically involving anonymous decisions and monetary payoffs, may fail to engage learners or to connect with their existing knowledge and concerns. Moreover, the game's framing does not explicitly connect to real-world institutions such as taxation, leaving participants to make the connection themselves if they can.

Tax-opoly addresses these limitations by embedding the core logic of the Public Goods Game within a concrete, thematic context. Players are not simply deciding how much to contribute to an abstract pool; they are deciding whether to pay taxes that fund visible public services such as schools, hospitals, and roads. This contextualisation makes the collective action dilemma tangible and personally meaningful, potentially enhancing both engagement and learning.

1.5.3 SimCity (Maxis, 1989)

SimCity, first released in 1989 and updated in multiple versions since, is a city-building simulation game that allows players to design and manage urban environments (Maxis, 1989). Players set tax rates, allocate budgets to public services such as police, fire, and education, and respond to challenges such as crime, pollution, and natural disasters. The game provides a rich, systems-level view of urban management and illustrates the relationships between tax revenue, public investment, and quality of life.

Research on SimCity's educational potential has generally been positive. Adams (1998) found that the game could effectively teach systems thinking and the interconnectedness of urban policies. However, the game's complexity and open-ended nature can also be overwhelming for new players, and its commercial orientation means that educational objectives are secondary to entertainment.

For the Tax-opoly project, SimCity offers valuable lessons about how to model the relationship between taxation and public goods. However, Tax-opoly's simpler, more focused design aims to make these concepts accessible in a shorter time frame and to players with no prior gaming experience. By reducing complexity while retaining core cause-and-effect relationships, Tax-opoly seeks to achieve educational goals that SimCity's broader scope may dilute.

1.5.4 Civilization Series (Sid Meier, 1991)

The Civilization series, beginning with Sid Meier's Civilization in 1991, is a turn-based strategy game that spans human history, from ancient times to the near future (Meier, 1991). Players guide a civilisation through technological, cultural, and political development, making decisions about research, governance, and resource allocation. Taxation appears as one element among many, with players setting tax rates that affect both immediate revenue and long-term growth.

Civilization's educational value has been widely recognised. Squire (2004) studied how the game could be used in classroom settings, finding that it engaged students with historical and geographical concepts and encouraged systems thinking. However, the game's enormous scope means that taxation is only one small part of a much larger system, and players may not focus on or remember the tax-related lessons amidst the many other demands on their attention.

Tax-opoly draws inspiration from Civilization's demonstration that complex societal systems can be modelled in engaging games. However, by focusing specifically on taxation and public goods, Tax-opoly aims to achieve deeper learning in this domain than a more general game could provide.

1.5.5 Taxland (OECD)

Taxland is an educational game developed by the Organisation for Economic Co-operation and Development to teach young people about taxation and civic responsibility (OECD, 2020). The game presents simplified scenarios in which players make decisions about tax compliance and see the consequences for their virtual community. Its development by a major international organisation signals growing recognition of the need for innovative tax education tools.

While Taxland represents an important initiative, its design targets relatively young learners (approximately ages 8-12) and may not engage older students or adults. The game's mechanics are relatively simple, and its strategic depth is limited. Tax-opoly aims to build on Taxland's foundation while targeting a broader age range especially that which includes university students (approximately ages 18-25).

1.5.6 Kahoot! (Gamified Learning Platform)

Kahoot! is a game-based learning platform that allows educators to create and deliver interactive quizzes (Kahoot, 2013). Players respond to questions using their devices, earning points for speed and accuracy. The platform's success, with billions of cumulative players, demonstrates the appeal of gamified learning and the effectiveness of simple competition in engaging learners.

However, Kahoot's model is fundamentally one of knowledge recall rather than experiential learning. Players answer questions about topics but do not experience simulated systems or make decisions with consequences. As Wang (2015) notes, while Kahoot can effectively review and reinforce existing knowledge, it may be less suited to teaching new, complex concepts that require deeper engagement.

Tax-opoly complements rather than competes with platforms like Kahoot!. While Kahoot can assess knowledge about taxation, Tax-opoly aims to build that knowledge through experiential learning. The two approaches could potentially be combined, with Kahoot quizzes used to reinforce lessons learned through Tax-opoly gameplay.

1.5.7 Budget Hero (Urban Institute & Brookings Institution, 2010)

Budget Hero was an online simulation game developed by the Urban Institute and Brookings Institution that allowed players to make decisions about the US federal budget and see the long-term consequences for debt, the economy, and various policy goals (Urban Institute & Brookings Institution, 2010). Players could increase or decrease spending in various categories and adjust tax policies, with the game providing immediate feedback on the fiscal and economic impacts.

Budget Hero represented a sophisticated approach to public finance education, making complex budgetary trade-offs accessible to general audiences. Its development by respected policy organisations demonstrates the potential of games to communicate serious policy content. However, the game focused on federal budgeting at a high level of aggregation, not on the individual decisions about tax compliance that are Tax-opoly's focus.

1.5.8 Foldit (University of Washington, 2011)

Foldit is a scientific puzzle game developed by the University of Washington that allows players to contribute to protein-folding research (University of Washington, 2011). Players manipulate protein structures in three dimensions, competing to achieve the most stable configurations. Remarkably, Foldit players have made genuine scientific discoveries, including solving the structure of a retroviral protease that had eluded researchers for years (Khatib et al., 2011).

Foldit's success demonstrates that games can not only teach about complex topics but also enable players to contribute meaningfully to scientific knowledge. While Tax-opoly does not aim for such ambitious outcomes, Foldit provides an inspiring example of what well-designed educational games can achieve. It also illustrates the power of making abstract, technical content tangible and manipulable through interactive gameplay.

1.6 LESSONS LEARNT FROM THE LITERATURE

The literature reviewed in this document yields several important lessons that inform the design and development of Tax-opoly:

First, tax compliance is a complex phenomenon shaped by economic, psychological, and social factors. While deterrence matters, so do trust, fairness perceptions, and understanding of how tax revenues are used. Educational interventions that address these factors have the potential to improve both attitudes toward taxation and actual compliance behaviour.

Second, the public goods nature of taxation is not intuitively understood by many citizens. The free rider problem and the benefits of collective investment are abstract concepts that benefit from concrete illustration. Games that allow players to experience the consequences of contribution decisions can make these concepts tangible and memorable.

Third, game-based learning and gamification, when well designed, can significantly enhance educational outcomes. Interactive, experiential learning environments engage learners more deeply than passive instruction and support better retention and transfer of knowledge. However, game mechanics must be thoughtfully aligned with learning objectives; poorly designed gamification can actually impede learning.

Fourth, existing games and applications provide both inspiration and cautionary tales. Monopoly demonstrates the enduring appeal of economic simulation but models only competition, not cooperation. The Public Goods Game captures the collective action dilemma but in an abstract form that may not engage general audiences. SimCity and Civilization illustrate how complex systems can be modelled in games but may be too broad and complex for focused educational objectives. Taxland shows that international organisations recognise the need for tax education games but leaves room for more sophisticated approaches.

Fifth, there is a gap in the current landscape for a game that: (a) focuses specifically on taxation and public goods; (b) targets a broad audience including young adults; (c) balances strategic depth with accessibility; (d) models both individual incentives and collective outcomes; and (e) can be easily deployed in educational settings. Tax-opoly is positioned to fill this gap.

1.7 CRITIQUE OF THE REVIEWED LITERATURE

While the literature provides a strong foundation for this project, several limitations and gaps deserve critical attention.

Theoretical limitations: Much of the tax compliance literature focuses on individual decision-making in developed country contexts. The applicability of these models to developing economies like Zambia, where formal institutions may be weaker and informal economic activity more prevalent, is not always clear. Research specifically examining tax morale and compliance behaviour in sub-Saharan African contexts remains relatively limited, though organisations like ActionAid Zambia (n.d.) and the Zambia Revenue Authority (2023) are beginning to address this gap.

Methodological concerns: Experimental studies of public goods and tax compliance, including those using the Public Goods Game, typically involve relatively small samples of university students in laboratory settings. The extent to which findings generalise to broader populations and real-world contexts is debated. Moreover, the abstract, decontextualised nature of many experiments may limit their relevance for understanding behaviour in complex, real-world situations.

Evidence gaps in game-based learning: While the overall evidence for game-based learning is positive, the literature suffers from several limitations. Many studies are small-scale and short-term, measuring immediate learning gains but not retention or behavioural change. Comparability across studies is limited by the diversity of games, contexts, and outcome measures. Relatively few studies have examined games specifically targeting tax or civic education, creating a gap that this project can help address.

Commercial versus educational priorities: The existing games reviewed in this document were designed primarily for entertainment or research rather than education. Even those with educational potential, such as SimCity and Civilization, were not designed with specific learning objectives in mind and may not optimise educational outcomes. Taxland, designed for education, sacrifices depth for accessibility. There remains space for a game that combines educational rigour with engaging gameplay.

Technological accessibility**:** Many existing educational games and simulations require significant computing resources or internet connectivity, limiting their accessibility in resource-constrained environments. For a project targeting contexts like Zambia, where connectivity and device availability may be limited, attention to low-tech and offline-capable design is essential. This consideration has informed Tax-opoly's design approach, which prioritises simplicity and broad compatibility.

1.8 CONCLUSION

This literature review has examined the theoretical and empirical foundations for developing Tax-opoly, an educational game designed to improve understanding of taxation and public goods. The review has drawn upon research in tax compliance, behavioural economics, public goods theory, game-based learning, and gamification, as well as analysis of existing applications relevant to the project's objectives.

Key findings from the literature include: (1) tax compliance is influenced by both economic deterrence factors and psychological factors such as trust, fairness perceptions, and tax morale; (2) the public goods nature of taxation is not intuitively understood, creating a need for educational interventions that make collective action dilemmas tangible; (3) well-designed game-based learning and gamification can significantly enhance educational outcomes by engaging learners in active, experiential problem-solving; (4) existing games and applications provide useful models but leave gaps that Tax-opoly can fill, particularly in combining focus on taxation with strategic depth and accessibility.

The literature also reveals important gaps: limited research on tax compliance in developing country contexts; methodological limitations in experimental studies; insufficient evidence on long-term outcomes of game-based learning; and a lack of games specifically designed to teach about taxation and public goods to broad audiences. Tax-opoly, while modest in scope as a university project, can contribute to addressing these gaps by demonstrating the feasibility and potential impact of a focused tax education game.

REFERENCES

ActionAid Zambia. (n.d.). *Tax justice and domestic resource mobilisation in Zambia*. ActionAid International.

Adams, P. C. (1998). Teaching and learning with SimCity 2000. *Journal of Geography*, 97(2), 47-55.

Allingham, M. G., & Sandmo, A. (1972). Income tax evasion: A theoretical analysis. *Journal of Public Economics*, 1(3-4), 323-338.

Alm, J., McClelland, G. H., & Schulze, W. D. (1992). Why do people pay taxes? *Journal of Public Economics*, 48(1), 21-38.

Alm, J., & Torgler, B. (2006). Culture differences and tax morale in the United States and in Europe. *Journal of Economic Psychology*, 27(2), 224-246.

Balaguer-Coll, M. T., Brun-Martos, M. I., Forte, A., & Tortosa-Ausina, E. (2019). Assessing public financial transparency: Evidence from local governments. *Government Information Quarterly*, 36(1), 101-116.

Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy*, 76(2), 169-217.

Commonwealth Secretariat. (2017). *Financial education for inclusive growth*. Commonwealth Secretariat.

CORE Economics Team. (2020). *The economy: Economics for a changing world*. Oxford University Press.

Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From game design elements to gamefulness: Defining "gamification". In *Proceedings of the 15th International Academic MindTrek Conference* (pp. 9-15).

Duolingo. (2023). *About Duolingo*. Duolingo, Inc.

Fehr, E., & Gächter, S. (2000). Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4), 980-994.

Gee, J. P. (2003). *What video games have to teach us about learning and literacy*. Palgrave Macmillan.

Hasbro. (2019). *Monopoly: Rules and gameplay overview*. Hasbro Inc.

Hsu, C. Y., Chang, C. T., & Hung, J. W. (2018). The influence of game-based learning on learning motivation and achievement. *Educational Technology & Society*, 21(2), 132-143.

Kahoot! (2013). *Kahoot! game-based learning platform*. Kahoot! AS.

Khatib, F., DiMaio, F., Cooper, S., Kazmierczyk, M., Gilski, M., Krzywda, S., ... & Baker, D. (2011). Crystal structure of a monomeric retroviral protease solved by protein folding game players. *Nature Structural & Molecular Biology*, 18(10), 1175-1177.

Kirchler, E. (2007). *The economic psychology of tax behaviour*. Cambridge University Press.

Ledyard, J. O. (1995). Public goods: A survey of experimental research. In J. H. Kagel & A. E. Roth (Eds.), *The handbook of experimental economics* (pp. 111-194). Princeton University Press.

Maxis. (1989). *SimCity* [Video game]. Electronic Arts.

Meier, S. (1991). *Sid Meier's Civilization* [Video game]. MicroProse.

Olson, M. (1965). *The logic of collective action: Public goods and the theory of groups*. Harvard University Press.

Organisation for Economic Co-operation and Development. (2020). *Taxland: Teaching taxes and civic responsibility*. OECD Publishing.

Pilon, M. (2015). *The Monopolists: Obsession, fury, and the scandal behind the world's favorite board game*. Bloomsbury.

Prensky, M. (2001). *Digital game-based learning*. McGraw-Hill.

Samuelson, P. A. (1954). The pure theory of public expenditure. *Review of Economics and Statistics*, 36(4), 387-389.

Seaborn, K., & Fels, D. I. (2015). Gamification in theory and action: A survey. *International Journal of Human-Computer Studies*, 74, 14-31.

Siregar, M., & Suryanto, T. (2024). Gamified learning environments in financial education. *Journal of Educational Technology*, 18(2), 45-58.

Squire, K. (2004). *Replaying history: Learning world history through playing Civilization III*. Indiana University.

Torgler, B. (2007). *Tax compliance and tax morale: A theoretical and empirical analysis*. Edward Elgar.

University of Washington. (2011). *Foldit: Collective problem-solving through games*. University of Washington.

Urban Institute & Brookings Institution. (2010). *Budget Hero: Understanding fiscal policy tradeoffs*. Urban Institute.

Vesselinov, R., & Grego, J. (2012). *Duolingo effectiveness study*. City University of New York.

Wang, A. I. (2015). The wear out effect of a game-based student response system. *Computers & Education*, 82, 217-227.

Wouters, P., van Nimwegen, C., van Oostendorp, H., & van der Spek, E. (2013). A meta-analysis of the cognitive and motivational effects of serious games. *Journal of Educational Psychology*, 105(2), 249-265.

Zambia Revenue Authority. (2023). *Tax compliance and revenue performance reports*. ZRA.